**University of Heidelberg**

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Bachelor's Thesis in Computer Science

# Data-driven camera calibration for light field camera systems

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# Abstract

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# 1. Introduction

## 1.1. Motivation

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The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

## 1.4. Structure of the Thesis

# 2. Theoretical Background

This chapter introduced the theoretical foundations required for this thesis. It covered the pinhole camera model, including intrinsic and extrinsic parameters as well as lens distortion, as the basis for geometric camera modeling. Building on this, common camera calibration methods and error metrics were presented, including target-based calibration and bundle adjustment. The chapter also introduced light field camera systems, their main system types, and the state of the art in their calibration. Together, these concepts provide the necessary background for the calibration approach and experiments presented in the following chapters.

## 2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [15].

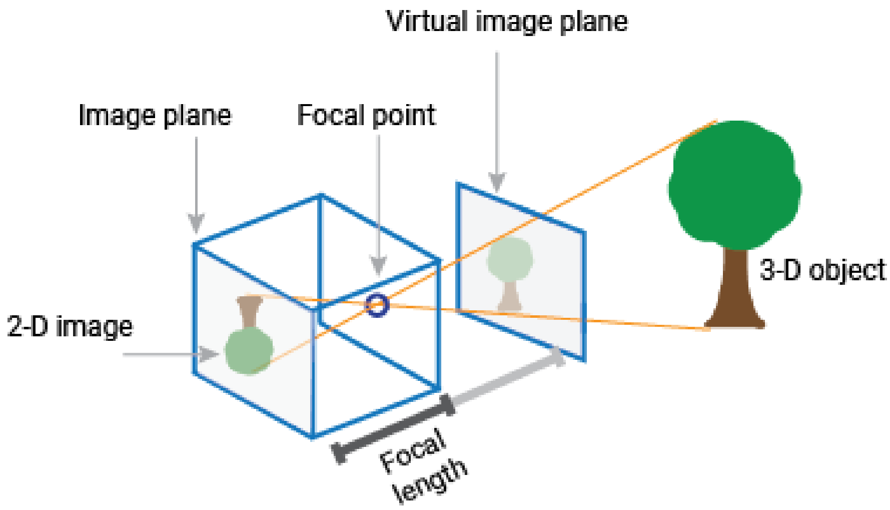


Figure 1: Illustration of the pinhole camera model. Source: [17].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [15]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [17].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

  where $(f_x, f_y)$ are the focal lengths in pixel units along the x and y axes, $s$ is the skew coefficient between the axes, and $(c_x, c_y)$ is the principal point (optical center) in pixel coordinates. $K$ is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

  where $R$ is a $3 \times 3$ rotation matrix representing the orientation of the camera with respect to the world coordinate system, and $t$ is a $3 \times 1$ translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

## 2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [17].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [6]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [20].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

### 2.2.1. Radial Distortion

Radial distortion is caused by the spherical shape of camera lenses, which leads to different refraction of light from the image center to the outer regions. This causes straight lines in the image to appear curved, resulting in barrel or pincushion distortion [5].

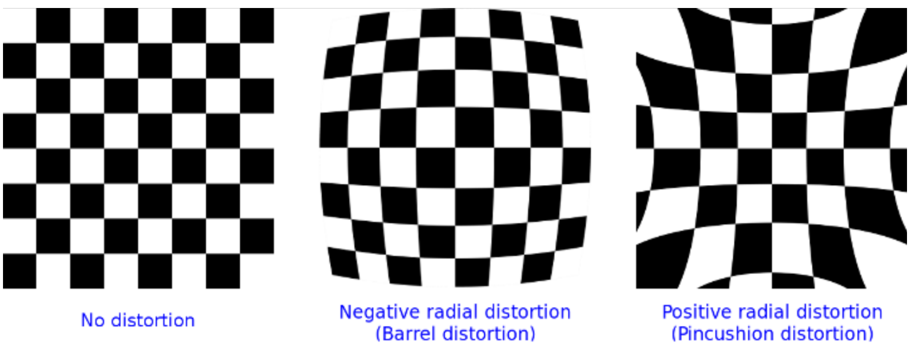


Figure 2: Comparison of different types of radial distortion. Source: [19].

The formal definition of radial distortion is given as follows [17]:

$$x_{distorted} = x\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) \tag{1}$$

$$y_{distorted} = y\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) \tag{2}$$

where $x$ and $y$ denote the undistorted normalized image coordinates in the camera coordinate system. The radius $r$ represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \tag{3}$$

The coefficients $k_1$, $k_2$, and $k_3$ in (1) and (2) control the strength of radial lens distortion. In practice, the first coefficient $k_1$ has the dominant influence on the distortion type: negative values of $k_1$ typically result in barrel distortion, while positive values lead to pincushion distortion.

### 2.2.2. Tangential Distortion

Tangential distortion occurs when the image sensor and the camera lens are not perfectly aligned or parallel to each other [17]. As a result, image points are displaced tangentially with respect to the optical axis. Figure 3 illustrates the geometric cause of tangential distortion and its effect on the captured image.

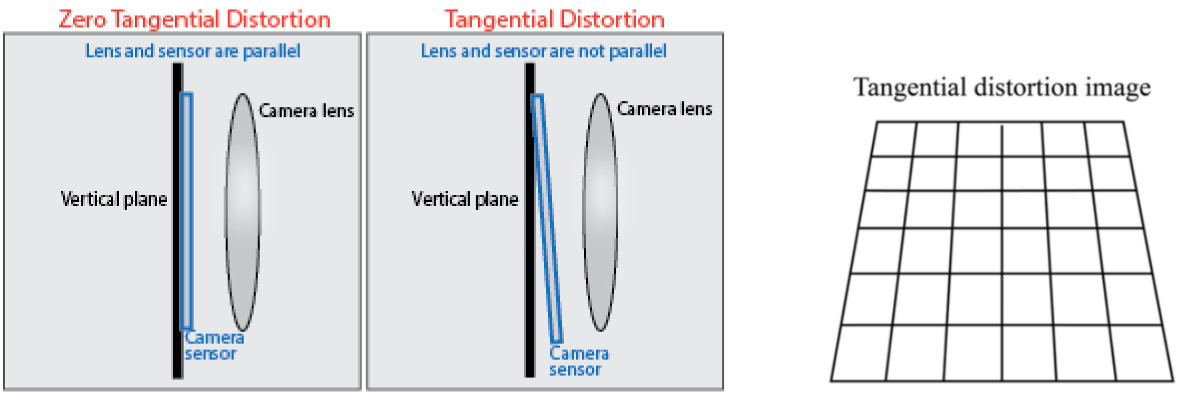


Figure 3: Visualization of tangential distortion. Source: [17], [16].

The formal definition of tangential distortion is given by the following equations [17]:

$$x_{\text{distorted}} = x + \left(2p_1xy + p_2\left(r^2 + 2x^2\right)\right) \tag{4}$$

$$y_{\text{distorted}} = y + \left(p_1\left(r^2 + 2y^2\right) + 2p_2xy\right) \tag{5}$$

Here, $x$ and $y$ denote the undistorted normalized image coordinates in the camera coordinate system. The radius $r$ represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \tag{6}$$

The coefficients $p_1$ and $p_2$ describe how strong the tangential lens distortion is. This type of distortion is caused by mechanical misalignment of the lens elements relative to the image sensor, resulting in a displacement of image points. Although tangential distortion is typically less pronounced than radial distortion, it can still negatively affect the camera calibration accuracy if not properly modeled.

## 2.3. Camera Calibration

The goal of camera calibration, also known as camera resectioning, is to estimate the parameters of a camera model. These parameters include the intrinsic parameters, the extrinsic parameters, and the lens distortion coefficients [17]. Knowing these parameters makes it possible to correct distortions in the recorded images and to perform metric measurements in the scene, for example to estimate object sizes or distances from image data.

Camera calibration is required for many tasks in 3D computer vision and photometry. Applications such as depth estimation, 3D reconstruction, or pose estimation rely on a precise mapping from 3D world coordinates to 2D image coordinates [21]. If the calibration is inaccurate, errors can propagate through the processing pipeline and reduce the quality of the final results.

In practice, cameras are often calibrated using targets with known geometric properties. Planar targets, such as checkerboard patterns, are widely used because they are easy to handle and lead to reliable results [28]. Several images of the target are captured from different viewpoints and orientations. Based on these images, the camera parameters can be estimated with relatively small calibration errors. For this reason, target-based calibration methods are commonly used in many computer vision applications.

### 2.3.1. Target-Based Camera Calibration

There are various approaches for target-based camera calibration. One of the most common methods uses a checkerboard pattern, as mentioned before. The checkerboard consists of a regular grid of black and white squares with known dimensions, as shown in Figure 4.

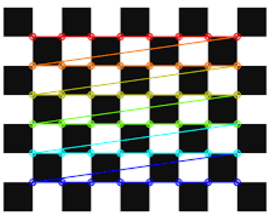

Figure 4: Example of a checkerboard and detected corner points. Source: [12].

During the calibration process, control points are extracted from the images. In the case of a checkerboard, these control points correspond to the corner points of the squares [21]. The positions of these corners are known in the coordinate system of the calibration target. By capturing multiple images of the checkerboard from different angles and distances, a set of 2D image points can be obtained that correspond to the known 3D points of the target. Based on these point correspondences, an optimization problem can be formulated to estimate the intrinsic and extrinsic camera parameters as well as the lens distortion coefficients [28].

This calibration approach is implemented in widely used computer vision libraries such as OpenCV [18]. It can be applied to single-camera setups and multi-camera systems. Besides checkerboards, other types of calibration targets exist, for example circular dot patterns [21]. In some cases, these patterns can achieve higher accuracy, especially when the image quality is low. However, they also require physical targets and a manual calibration setup.

The main limitations of target-based calibration methods are the required manual placement of the target and the need to capture images with different poses. In addition, the calibration accuracy depends on factors such as the quality of the corner detection, the number of images used, and the diversity of viewpoints during data acquisition [28]. Despite these limitations, target-based calibration remains a standard method in many computer vision applications due to its simplicity and reliable performance.

#### 2.3.2. Calibration Parameters and Error Metrics

To evaluate the quality of a camera calibration, different error metrics are used. These metrics describe how well the estimated camera parameters fit the observed image data. In practice, this is done by comparing detected image points with points that are projected using the calibrated camera model. The most common metric is the reprojection error [17].

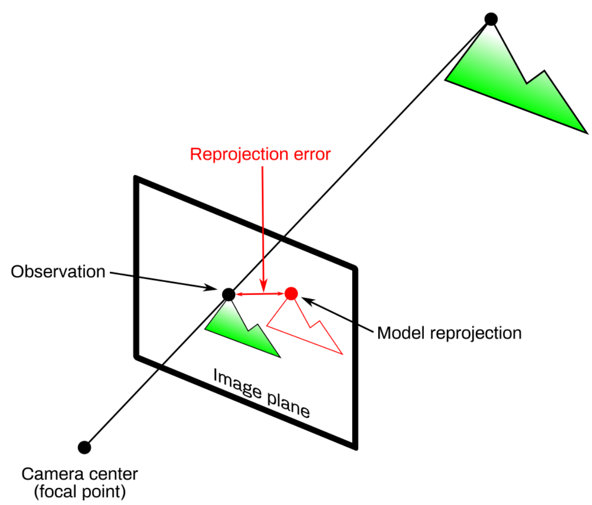


Figure 5: Visualization of reprojection error. Source: [2].

After a calibration, known 3D points are projected back onto the image plane. The reprojection error measures the distance between the observed image points and the corresponding projected points. It is usually computed as the average Euclidean distance and is given in pixel units [2]. It can be defined as

$$e = \frac{1}{N} \sum_{i=0}^{N-1} \|\mathbf{p}_i - \mathbf{q}_i\|_2. \tag{7}$$

Here, $\mathbf{p}_i$ represents the observed feature points in the image, while $\mathbf{q}_i$ denotes the projected points obtained from the calibrated camera model. In general, a smaller reprojection error indicates a better calibration. What is considered a good value depends on the application, but for accurate calibrations the reprojection error is often between 0.1 and 0.5 pixels [11].

Another commonly used measure is the root mean square (RMS) error [10]. It summarizes the reprojection error over all calibration points and is computed as:

$$\epsilon_{\text{res}} = \left( \frac{1}{2n} \sum_{i=1}^{n} \left\| \mathbf{p}_i' - \mathbf{q}_i' \right\|_2^2 \right)^{1/2} \tag{8}$$

The RMS error provides a single value that describes the overall calibration accuracy and is often used to compare different calibration results.

### 2.3.3. Bundle Adjustment

Bundle adjustment is a method for improving a 3D reconstruction by refining both the positions of 3D points and the camera parameters at the same time [23]. The name comes from the “bundles” of light rays from the 3D points to the cameras, which are adjusted together to make the reconstruction more accurate. The parameters that are optimized together usually include the 3D coordinates of the points, the extrinsic parameters, and sometimes the intrinsic parameters [3].

This joint optimization improves calibration accuracy because it ensures that all cameras and points are consistent with each other. As a result, projection errors are reduced and small misalignments are corrected that would remain if cameras or points were adjusted separately [23]. This idea is illustrated in Figure 6.

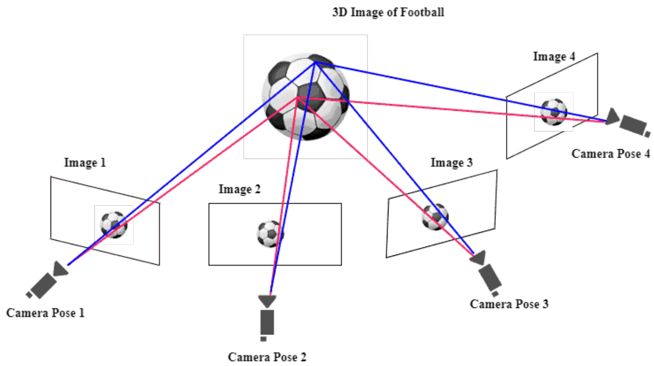


Figure 6: Visual representation of bundle adjustment. Source: [13].

Mathematically, bundle adjustment is formulated as an optimization problem that minimizes the reprojection error. This optimization is usually solved using nonlinear least squares methods and takes advantage of the fact that the problem structure is largely sparse [3].

## 2.4. Light Field Camera Systems

A standard camera captures a 2D projection of the 3D world. Light field cameras, on the other hand, have the ability to capture more information. They record not only the intensity of light rays but also their direction [9]. This property allows the creation of accurate depth maps and refocus the image after it has been taken [7]. Therefore light field cameras are growing in importance for applications in computer vision, robotics, augmented reality, and other applications.

The four-dimensional nature of light fields means that their processing currently needs relatively high computational resources [9]. However, the evolution of hardware and algorithms is making light field cameras more practical for real-world applications.

There are mainly two different types of light field camera systems: multi-camera arrays and micro-lens-array-based cameras. Each type has its own advantages and disadvantages, which will be discussed in the following sections.

### 2.4.1. Multi-Camera Light Field Systems

Multi-camera arrays are one way to capture light fields. This type of system is structured by arranging multiple individual cameras in a geometric configuration [25]. Each camera captures a different viewpoint of the scene, enabling the system to record a 4D light field. In practice, there are different approaches to arranging the cameras. For example, in [22], an array of 128 cameras in a square grid is used, while in other works cross-shaped structures with fewer cameras are employed [25], as shown in Figure 7.

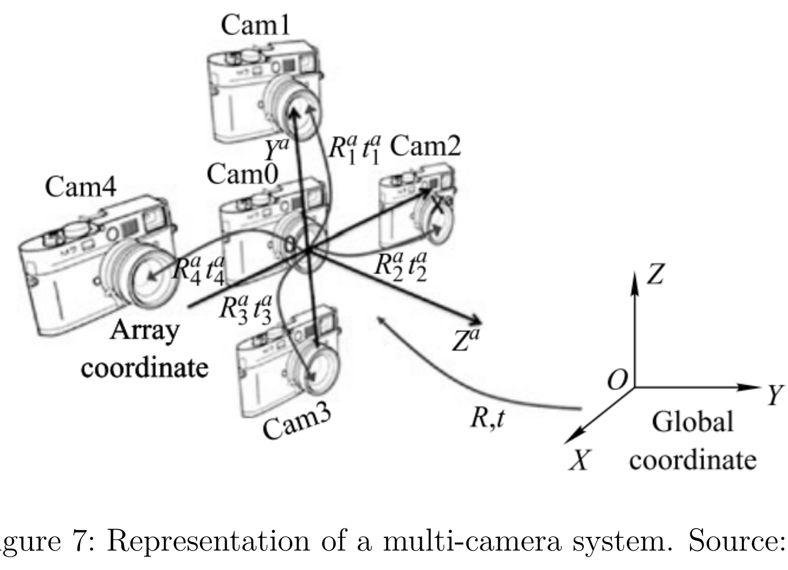


Figure 7: Representation of a multi-camera system. Source: [25].

Besides the advantages of a multi-camera setup, such as improved depth accuracy and refocusing capabilities, there are also challenges. The calibration of such systems is more complex and requires higher computational effort due to the larger number of cameras involved [25].

It is also essential to keep the cameras properly aligned and synchronized. Even small errors in alignment or timing can cause artifacts or undesired results in the final image. Therefore, multi-camera light field systems require a very precise setup and calibration to work as intended.

### 2.4.2. Micro-Lens-Array-Based Light Field Cameras

The second type of light field camera is based on a micro-lens array (MLA). In this case instead of having different cameras, the system uses only one sensor combined with an array of small lensen placed between the main lens and the image sensor [7]. The MLA allows the camera to capture also the 4D light field information.

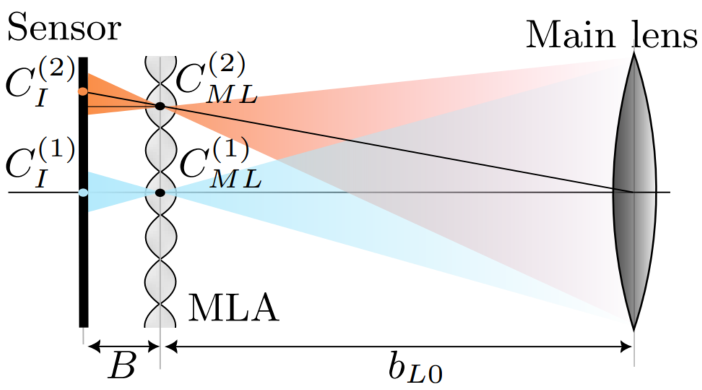


Figure 8: Structure of a micro-lens-array-based light field camera. Source: [7].

As illustrated in Figure 8, each lens in the MLA focuses light from different directions onto different regions of the sensor. This way, the camera is also able to cpature the same information as a multi-camera array, but with a more compact design. Because of the smaller size and lower cost, MLA-based light field camera are mostly used in consumer applications [14]. However, microlens-array cameras have to compromise between spatial and angular resolution, while multi-camera systems capture full-resolution images with sharper results and more accurate depth information.

For MLA-based light field cameras, the pattern-based calibration method provides satisfactory results in terms of reprojection error. In practice, the overall calibration quality strongly depends on the calibration target, particularly its flatness [1].

## 2.5. State of the Art in Light Field Camera Calibration

Accurate calibration is a central requirement for light field camera systems, as even small geometric errors directly affect depth estimation and view consistency. In camera arrays, where many individual views must agree, calibration errors accumulate and lead to visible artifacts in depth maps and reconstructed images.

Early work on light field calibration mainly relies on pattern-based approaches. Dima et al. systematically evaluate several multi-camera calibration methods using ground truth data and show that calibration accuracy strongly depends on the array geometry and the physical setup of the calibration target [4]. In particular, small deviations such as bending or uneven surfaces of the calibration board already lead to increased reprojection errors across the array. Yang et al. demonstrate that using a simple checkerboard target can still yield stable calibration results for camera arrays while reducing setup complexity [26]. However, the accuracy of these methods remains closely tied to target quality and careful placement.

To overcome the limitations of target-based calibration, several self-calibration approaches have been proposed. Zhang et al. introduce a fixed relative pose prior to stabilize self-calibration in camera arrays [27]. By constraining relative camera poses, their method reduces pose drift in situations where image features are sparse or unevenly distributed. Zhou et al. address low-texture sports scenes and use temporal fusion across multiple frames to estimate calibration parameters [29]. These approaches reduce the need for calibration targets, but still depend on sufficient scene structure and a reasonable initial alignment.

Some methods focus on specific system layouts or downstream tasks. Ge et al. study a ring-shaped multi-camera system and use a transparent calibration target to support panoramic measurements [8]. Xie et al. show that accurate camera array calibration is essential for reliable depth map fusion, as calibration errors propagate directly into depth artifacts, especially near object boundaries [24].

The approach most relevant to this thesis is LiFCal by Fleith et al. [7]. Instead of using a physical calibration target, LiFCal moves the MLA-based camera through the scene and estimates calibration parameters via bundle adjustment driven by depth maps from multiple viewpoints. This shifts the calibration problem from controlled target acquisition to scene structure and camera motion, making the method particularly suitable for scenarios where calibration targets are impractical or unavailable.

Overall, existing work shows that calibration accuracy in light field systems remains sensitive to setup quality, motion, and scene structure. Larger camera arrays amplify small errors, and target-free methods require reliable depth estimates and sufficient camera motion to properly constrain the geometry.

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## 6.1. Conclusion

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# Acknowledgment

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